

Contoso Precision Parts — Synthetic Corpus Readme

Participant guide to the 21-PDF dataset for Microsoft Foundry IQ Hackathon Challenge 2. Read this first.

```
--- METADATA ---
document_id: DATA-README
title: Contoso Precision Parts – Synthetic Corpus Readme
domain: readme
source_system: Hackathon
classification: public
revision: Rev A
effective_date: 2026-05-18
owner: Hackathon facilitator – FoundryIQ Challenge 2
part_numbers: []
operation_numbers: []
machine_ids: []
controller_models: []
fault_codes: []
materials: []
standards: []
supplier_ids: []
jurisdictions: []
classification_note: Safe to share with all participants
---
```

1. What this dataset is

Contoso Precision Parts is a fictional mid-sized precision-machining shop in Wichita, KS, that supplies parts to aerospace (Aeronova), automotive (Vertex Motors), and industrial customers. The 21 PDFs in this folder are a *synthetic but internally consistent* knowledge corpus that lets you build and demo a Foundry IQ agent against a realistic manufacturing scenario without using real customer data.

The corpus is intentionally designed so that the most interesting questions **cannot be answered from a single document**. They require the agent to combine an OEM machine manual with a quality control plan with a process spec — exactly the cross-source reasoning that the hackathon judging rubric rewards (see §6).

Single source of truth: every machine ID, part number, operation number, fault code, material code, standard, and supplier ID used in any PDF is declared once in `ENTITY-BACKBONE.md`. If you generate additional documents, add their entities there first.

2. How the corpus is organised

Files are grouped into five domain folders that mirror the five source systems an indexer would crawl in a real Contoso environment:

Folder	Domain	Source system	Files
01-machine-docs/	Machine documentation	PDM / PLM	4
02-quality/	Quality standards	QMS	6

03-process-specs/	Process specifications	PLM	4
04-materials/	Materials & supplier data	ERP	4
05-compliance/	Regulatory compliance	SharePoint	3

Every PDF carries a structured `--- METADATA ---` block on page 1. Those fields (`document_id` , `revision` , `classification` , `part_numbers` , `machine_ids` , `fault_codes` , `materials` , `standards` , `supplier_ids` , `effective_date` , `safety_critical`) are the filterable facets your Azure AI Search index should lift. Filterable metadata is weighted heavily in the rubric — do not skip it.

3. The anchor entities (memorise these)

Almost every cross-document question pivots on one of these:

Kind	Anchor IDs	Why it matters
Machines	DMU-14 , DMU-22 (DMG MORI 5-axis); NLX-08 , NLX-09 (Mori NLX-2500 turning); HMC-30 ; BSAW-03 (band saw)	All four CNC machines run the Fanuc Oi-TF controller, so alarm codes are shared across the fleet.
Controller	Fanuc Oi-TF	Alarm 300 = position-encoder fault. Operators sometimes mis-attribute it to tool wear — the corpus deliberately contradicts that guess.
Parts	45621-B (Aeronova turbine bracket, Inconel 718); 34521-A (Vertex steering knuckle, 4140 steel); 78901-C (industrial housing, GG25 cast iron)	The first two are the hardest to manufacture and have the deepest documentation trail.
Operations	OP10 saw-cut → OP20 rough → OP30 finish → OP40 inspect	OP30 is where the customer-critical tolerances are produced.
Materials	MAT-INC718 , MAT-4140 , MAT-GG25 , MAT-CF-ECO (EcoCut 7700 coolant), MAT-WD40	Material code links the part, the process spec, the SDS, and the approved supplier.
Standards	IATF-16949-2016 , AS9100D , 29-CFR-1910.147 (LOTO), 29-CFR-1910.212 (machine guarding), GHS-Rev9	Compliance questions cite the clause; your retrieval must surface the right clause excerpt, not just the standard name.

4. The 21 files — what's in each and what it relates to

4.1 Machine documentation (01-machine-docs/)

Document ID	What's in it	Relates to
PDM-MAN-FANUC-0iTF-001	Fanuc Oi-TF controller manual excerpt: alarm code reference, diagnostics workflow, recovery procedures. Definitive source for alarm 300 (position-encoder fault).	Applies to DMU-14/22 and NLX-08/09 (all run Fanuc Oi-TF). Cross-cited by PDM-SB-DMU14-2024-03 , PLM-PS-INC718-0042 , QMS-FMEA-45621B-RevA .
PDM-MAN-DMU14-007	DMG MORI DMU-14 5-axis OEM manual excerpt: spindle envelope, axis limits, max RPM (12,000), permissible tooling.	Hard upper bounds for PLM-WI-45621B-OP30-RevC — when the work instruction and OEM conflict, OEM wins.

PDM-MAN-NLX-004	Mori NLX-2500 turning centre manual: chuck, tailstock, live-tool module, turning envelope.	Bounds for PLM-WI-34521A-OP30-RevD (34521-A turning).
PDM-SB-DMU14-2024-03	DMG MORI service bulletin: B-axis encoder drift on DMU-14 fleet, calibration procedure, retrofit kit P/N.	Refines PDM-MAN-FANUC-01TF-001 alarm 300 guidance for the DMU-14 specifically.

4.2 Quality (02-quality/)

Document ID	What's in it	Relates to
QMS-IATF-EXCERPT-001	IATF 16949:2016 clause excerpts that drive Contoso quality procedures: §8.5.1.1 control-plan review cadence, §10.2 nonconformity & corrective action.	Cited by every control plan and FMEA. The authority document for Q2 and Q8.
QMS-CP-34521A-RevC	Current control plan for steering knuckle 34521-A: characteristics, tolerances, gauges, sample sizes, reaction plans per failure mode.	Linked to FMEA QMS-FMEA-34521A-RevB (failure modes FM-007 etc.) and work instruction PLM-WI-34521A-OP30-RevD .
QMS-CP-34521A-RevB	SUPERSEDED (Rev B, dated 2024-02-15). Kept on purpose: lets you demo revision-enforcement governance (agent must refuse to cite it as authoritative).	Superseded by Rev C. Same part, older inspection plan.
QMS-FMEA-34521A-RevB	FMEA for 34521-A: failure modes (FM-001...FM-010 incl. FM-007 bore-diameter out of tolerance), Severity/Occurrence/Detection, RPN, recommended actions.	Failure modes are referenced by ID in QMS-CP-34521A-RevC reaction plans.
QMS-FMEA-45621B-RevA	FMEA for turbine bracket 45621-B: includes FM-104 (tool-wear-driven surface degradation on Inconel finishing).	Cross-cites PLM-PS-INC718-0042 for tool-wear limits and PDM-MAN-FANUC-01TF-001 for alarm context.
QMS-CSR-VTX-001	Customer-specific requirements matrix from Vertex Motors. Classification: restricted — RBAC demo target.	Overlays QMS-CP-34521A-RevC : any conflict, customer spec wins.

4.3 Process specs (03-process-specs/)

Document ID	What's in it	Relates to
PLM-WI-45621B-OP30-RevC	Operator work instruction for 45621-B OP30 finishing on DMU-14: spindle RPM, feed, DOC, tool list, in-process checks.	Must stay within OEM bounds from PDM-MAN-DMU14-007 ; tool data comes from PLM-T00L-45621B-OP30 ; material rules from PLM-PS-INC718-0042 .
PLM-WI-34521A-OP30-RevD	Work instruction for 34521-A OP30 turning on NLX-08/09.	Bounded by PDM-MAN-NLX-004 ; characteristics flow from QMS-CP-34521A-RevC .
PLM-PS-INC718-0042	Process spec for machining Inconel 718: surface-speed/feed/DOC windows, VBmax = 0.20 mm tool-wear limit, coolant rules.	Authority for the tool-wear answer in Q7 (overrides operator's alarm-300 guess).

PLM-TOOL-45621B-OP30	Tool list for 45621-B OP30: insert grades, holders, gauge lengths, expected life.	Referenced by PLM-WI-45621B-OP30-RevC .
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4.4 Materials & suppliers (04-materials/)

Document ID	What's in it	Relates to
ERP-MAT-INC718-SPEC	Material specification for Inconel 718: chemistry, mechanical props, heat-treat condition, required certs (MTR, Nadcap) , approved suppliers.	Linked to part 45621-B , to PLM-PS-INC718-0042 , and to supplier entry in ERP-ASL-2025-Q3 .
ERP-SDS-ECOCUT7700	Safety Data Sheet for EcoCut 7700 coolant: GHS hazard codes incl. H315 (skin irritation) , PPE, first aid.	Used on DMU/NLX/HMC/BSAW machines — referenced by SP-OSHA-MG-WICHITA and SP-LOTO-BSAW03-RevB .
ERP-SDS-WD40-SPEC	SDS + spec for WD-40 (incidental lubricant).	Minor; included so SDS retrieval has more than one candidate.
ERP-ASL-2025-Q3	Approved supplier list, Q3 2025. Classification: restricted.	Authoritative cross-reference for material → approved supplier questions; covers all MAT-* codes and SUP-* IDs used elsewhere.

4.5 Compliance (05-compliance/)

Document ID	What's in it	Relates to
SP-OSHA-MG-WICHITA	OSHA machine-guarding summary for the Wichita facility: which 29 CFR 1910 clauses apply to which machines.	Cites 1910.212 (guarding) and 1910.147 (LOTO); points to SP-LOTO-BSAW03-RevB for band-saw.
SP-LOTO-BSAW03-RevB	Lock-out / tag-out procedure for the BSAW-03 band saw. safety_critical: true.	Must be cited together with ERP-SDS-ECOCUT7700 whenever the agent is asked about band-saw maintenance.
SP-PRODCERT-ANV-45621B	Aeronova-specific product certification package for 45621-B. Classification: restricted.	Closes the loop: part 45621-B → AS9100D → MTR/Nadcap → approved supplier.

5. How everything cross-references (mental model)

Read top-to-bottom; arrows mean "is constrained by / cites":

```

Part (45621-B)
├─ Customer requirements — QMS-CSR-VTX-001 (for 34521-A) / SP-PRODCERT-ANV-45621B (for 45621-B)
├─ Control plan — QMS-CP-{part}-Rev{X} ← supersedes Rev{X-1}
│   └─ Failure modes — QMS-FMEA-{part}-Rev{Y} (FM-### IDs)
├─ Work instruction — PLM-WI-{part}-OP30-Rev{Z}
│   ├── bounded by — PDM-MAN-{machine}-### (OEM hard limits)
│   ├── tool data — PLM-TOOL-{part}-OP30
│   └─ material rules — PLM-PS-{material}-####
├─ Material spec — ERP-MAT-{material}-SPEC
│   └─ Approved supplier — ERP-ASL-2025-Q3 (SUP-### IDs)
├─ Coolant / SDS — ERP-SDS-{product}
└─ Compliance — SP-OSHA-MG-WICHITA → SP-LOTO-{machine}-Rev{Z}

```

Authority always escalates: Customer CSR > Standard (IATF/AS9100) > OEM manual > Internal work instruction.
Revision always wins: newer Rev letter supersedes older. SUPERSEDED docs must not be cited as authoritative.

6. How to use this dataset for the hackathon

The corpus is built so each of the eight sample questions in the challenge pack requires the agent to retrieve from **two or three distinct documents in different folders** and reconcile them.

Q#	Question theme	Documents the agent must combine
Q1	What does Fanuc alarm 300 mean and how do I clear it?	PDM-MAN-FANUC-0iTF-001
Q2	How often must a control plan be reviewed under IATF 16949?	QMS-IATF-EXCERPT-001
Q3	What are the OP30 parameters for 45621-B and are they within the DMU-14's OEM limits?	PLM-WI-45621B-OP30-RevC + PDM-MAN-DMU14-007
Q4	How often is the bore on 34521-A inspected and what failure mode does that catch?	QMS-CP-34521A-RevC + QMS-FMEA-34521A-RevB
Q5	A bore on 34521-A came out oversize — what is the reaction plan?	QMS-CP-34521A-RevC + QMS-FMEA-34521A-RevB + pointer to PLM-WI-34521A-OP30-RevD
Q6	What PPE and lock-out steps apply to maintaining the band saw?	SP-OSHA-MG-WICHITA + SP-LOTO-BSAW03-RevB + ERP-SDS-ECOCUT7700
Q7	DMU-14 threw alarm 300 mid-cut on Inconel — is this tool wear?	PDM-MAN-FANUC-0iTF-001 + PLM-PS-INC718-0042 + QMS-FMEA-45621B-RevA (corpus deliberately contradicts the operator's guess — alarm 300 is encoder, not tool wear)
Q8	A 45621-B lot failed final inspection — what does IATF require and what cert does the material need?	QMS-IATF-EXCERPT-001 + QMS-CP-34521A-RevC + ERP-MAT-INC718-SPEC

Recommended build order

1. **Index everything** in Azure AI Search. Lift the YAML metadata block as filterable facets — *not* just as text. The rubric weights filterable metadata 30%+.
2. **Knowledge bases:** group documents by domain folder. Use the source-system facet to scope queries (e.g., compliance questions should not retrieve from PLM).
3. **Agentic retrieval:** wire up Foundry Agent Service with MCP. Each sample question above is a sub-query decomposition test. Trace the sub-queries — judges want to see the plan.
4. **Demo one governance control**, picking the easiest win from the four hooks already baked in (see §7).

7. Governance hooks baked in (pick at least one to demo)

Hook	Where in the corpus	Demo idea
Revision enforcement	QMS-CP-34521A-RevB is marked SUPERSEDED; Rev C is current.	Agent must refuse to cite Rev B as authoritative; cite Rev C and say what changed.
RBAC / classification	Three documents tagged classification: restricted : QMS-CSR-VTX-001 , ERP-ASL-2025-Q3 , SP-PRODCERT-ANV-45621B .	Show the agent withholding restricted content for an unauthorised persona; surfacing it for a quality-lead persona.

Staleness detection	QMS-CP-34521A-RevB dated 2024-02-15 is past its 12-month review window per IATF §8.5.1.1.	Agent flags "this document is overdue for review" before answering.
Safety-critical prompting	SP-LOT0-BSAW03-RevB and SP-OSHA-MG-WICHITA carry <code>safety_critical: true</code> .	Any answer that touches a safety-critical doc must include a mandatory safety reminder and a citation.

8. Metadata schema cheat sheet

Every PDF has a block on page 1 that looks like:

```

--- METADATA ---
document_id: PDM-MAN-FANUC-0iTF-001
title: Fanuc 0i-TF Operator Manual – Alarms & Diagnostics
domain: machine_docs
source_system: PDM
revision: Rev A
effective_date: 2024-06-01
owner: M. Tanaka, Process Engineering – Wichita
part_numbers: []
operation_numbers: []
machine_ids: [DMU-14, DMU-22, NLX-08, NLX-09]
controller_models: [Fanuc 0i-TF]
fault_codes: [300, 401, 506]
materials: []
standards: []
supplier_ids: []
jurisdictions: []
classification: internal      # public | internal | restricted
safety_critical: false
supersedes: PDM-MAN-FANUC-0iTF-000 (only if applicable)
---
```

For indexing, parse this block (regex: `---\s*METADATA\s*---(.+?)---` with DOTALL) and lift each key as either a string field or a multi-value collection field. List fields use `[a, b, c]` syntax — split on commas.

9. Regenerating & validating the corpus

If you need to extend or rebuild the dataset:

1. **Add the entity** to `ENTITY-BACKBONE.md` first. Don't invent new machine IDs / part numbers inside a PDF.
2. **Add a `register({...})` block** in `tools/generate_pdfs.py`. Follow the dict shape of any existing document.
3. **Run the generator:** `.\.venv\Scripts\python.exe tools\generate_pdfs.py` (regenerates all) or pass document IDs to regenerate a subset.
4. **Run the validator:** `.\.venv\Scripts\python.exe tools\verify_pdfs.py`. It round-trips every PDF through pypdf and asserts that every entity declared in metadata is actually present in the body text, and that the eight sample questions still have term-level coverage.
5. To regenerate just this readme: `.\.venv\Scripts\python.exe tools\generate_data_readme.py`.

Do not edit PDFs directly. They are generated artefacts. All content lives in the Python source so the corpus stays internally consistent.

10. What's intentionally *not* in the corpus

- **No real customer data.** Aeronova and Vertex Motors are fictional. Any resemblance to real OEM part numbers is coincidental.
- **No CAD or drawing files.** Only the textual artefacts Foundry IQ retrieval needs.
- **No ERP transactional data** (POs, receipts, lots). The ASL is the only ERP-system extract.
- **No video, images, or audio.** Pure text-PDF on purpose.
- **No multilingual content.** All English.

If a participant needs richer data (e.g., to demo multi-modal retrieval), extend the corpus following §9 — but please keep the ENTITY-BACKBONE invariants intact so the eight reference questions still work.

Document control / sign-off

Role	Name	Signature	Date
Document Owner	Hackathon facilitator — FoundryIQ Challenge 2	(on file)	2026-05-18
Approver	Hackathon facilitator — FoundryIQ Challenge 2	(on file)	2026-05-18